

| Assessment | Test | Domain        | Skill                                                                         | Difficulty                                          |
|------------|------|---------------|-------------------------------------------------------------------------------|-----------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear equations in one variable and systems of equations in two variables | Medium <div><div></div><div></div><div></div></div> |

Question

$$y = (x - 2)(x + 4)$$
$$y = 6x - 12$$

Which ordered pair  $(x, y)$  is the solution to the given system of equations?

Answer

- A.  $(0, 2)$
- B.  $(-4, 2)$
- C.  $(2, 0)$
- D.  $(2, -4)$

Correct Answer: C

Rationale

Choice C is correct. The second equation in the given system of equations is  $y = 6x - 12$ . Substituting  $6x - 12$  for  $y$  in the first equation of the given system yields  $6x - 12 = (x - 2)(x + 4)$ . Factoring 6 out of the left-hand side of this equation yields  $6(x - 2) = (x - 2)(x + 4)$ . An expression with a factor of the form  $(x - a)$  is equal to zero when  $x = a$ . Each side of this equation has a factor of  $(x - 2)$ , so each side of the equation is equal to zero when  $x = 2$ . Substituting 2 for  $x$  into the equation  $6(x - 2) = (x - 2)(x + 4)$  yields  $6(2 - 2) = (2 - 2)(2 + 4)$ , or  $0 = 0$ , which is true. Substituting 2 for  $x$  into the second equation in the given system of equations yields  $y = 6(2) - 12$ , or  $y = 0$ . Therefore, the solution to the system of equations is the ordered pair  $(2, 0)$ .

Choice A is incorrect and may result from switching the order of the solutions for  $x$  and  $y$ .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.